Handwritten Fields: Usable Information and Related Research

Finaira Privacy Gateway project, October 2026

How to use this document

The gateway does not read handwriting in general. It cuts each handwritten field out of the page, keeps it inside the organization, and puts it back after the external service answers. Some handwritten fields are useful and low in sensitivity, so they can be read locally and shared. This document lists those fields and the research related to this approach.

1. **Part 1** lists the handwritten fields that can be read locally, how they look on the page, and what may be shared. Priority labels (P0, P1, P2, Keep) follow the document *Financial Data Protection Rules and Sensitive Information List*.
2. **Part 2** lists papers and projects that crop parts of documents, mask them, or restore them.
3. **References** are at the end.

Part 1. Handwritten information that can be understood

A. Read and send the value (P2 and Keep)

Information	Priority	Where	How it looks	How to read locally
Gender	P2	KYC forms	Two ticks (ذكر / أنثى) or a word on a dotted line	Tick state, or two possible words. Also contained in the national ID (13th digit) if read locally.
Nationality, other nationalities	P2	KYC forms	A word on a line (مصري), sometimes an "Egyptian / other" tick	Tick, or a fixed list of countries
Marital status	P2	KYC forms and updates	Three or four ticks (أعزب، متزوج، مطلق، أرمل)	Tick
Education	P2	KYC forms	Four or five ticks (أمي، مؤهل متوسط، عالي، دراسات عليا)	Tick
Housing type	P2	KYC forms	Ticks (ملك، إيجار، أخرى)	Tick
Residency status	P2	KYC forms	Ticks (مقيم / غير مقيم)	Tick
Customer type	P2	KYC forms	Ticks	Tick
Branch name	P2	Most bank forms	A word on a line at the top	Match against the bank's branch list. Keep the request number masked: branch, request number and date together point to one record.
Statement frequency, method, language	Keep	Account opening	Ticks	Tick

Information	Priority	Where	How it looks	How to read locally
Account type, account currency	Keep	Account opening, deposit forms	Ticks	Tick
Transfer type	Keep	Paper transfer forms	Ticks (internal, wire, manager's cheque)	Tick
Charges option	Keep	Transfer forms	Three ticks (sender, beneficiary, shared)	Tick
Dispute reason	Keep	Card dispute forms	A column of 8 to 10 printed reasons with ticks	Tick
Card type requested	Keep	Card applications	Ticks	Tick
Filled or empty, signed or not, stamped or not, thumbprint present	Keep (derived)	Every form	Ink in a zone; a signature line; a round or rectangular stamp	Ink count per zone, no reading needed

B. Read but send only in a general form (P1)

Information	How it looks	Read locally as	Send
Date of birth	Digits in 6 to 8 boxes, or three blanks for day, month and year	Single digits	Age band
ID expiry date	Digits in boxes	Single digits	Valid or expired
Employment type	Ticks (قطاع عام، قطاع خاص، أعمال حرة، طالب، بالمعاش، لا يعمل، قوات مسلحة، شرطة)	Tick	Sector. Armed forces and police need a compliance decision.
Job title	Free words on a line	Hard (free handwriting)	Occupation category, only if read reliably; otherwise masked
Employment start date	Digits	Single digits	Years in job band
Politically exposed person	Yes or no tick	Tick	Yes or no, if policy allows
Place of birth	A word on a line	Governorate and country list	Country, or Egypt only

C. Borderline (compliance decision)

Loan purpose ticks, instalment frequency ticks, and the currency on a transfer. These may count as transaction or loan details under Law 194 Article 140.

Key point. Almost everything usable is a tick box or a single digit in a box. None of it needs Arabic handwriting recognition, and free handwriting is almost all P0 or P1 anyway. This supports the crop and restore design.

Part 2. Research on cropping and restoring document parts

No single work combines all four steps: cropping by layout, a typed placeholder, a cloud model round trip, and pasting the crop back. Each half exists.

Cropping fields from forms with a known layout

Work	What it does	Relevance
ODK Scan / ODK-X Scan (University of Washington, open source)	A JSON template stores each field's location. Bubbles, checkboxes and written numbers are read automatically, and image snippets of handwritten text are saved for manual entry.	Closest match to tiers A and B
Dell, Breit, Wobbrock, Borriello. "Improving Form-Based Data Entry with Image Snippets." Graphics Interface 2013	Segments a form image into one snippet per field. Data entry was 28.3 percent faster on desktop with no detectable loss of accuracy.	Academic evidence for field cropping
Shreddr (Chen, Kannan, Yano, Hellerstein, Parikh. ACM DEV 2012; later Captricity)	Cuts forms into field images for crowd workers. Over one million values were entered from 36,819 pages in Mali.	Same cropping at scale; workers see only pieces out of context
Snippet keying in commercial capture (IBM Datacap; ScaleHub snippeting; patent US 2015/0213283)	Operators see only one field's crop, never the full page, explicitly for privacy	Direct precedent for cropping as protection

Masking fixed regions (no restore)

Work	What it does
Laserfiche Fixed Annotation, Hyland OnBase, Tungsten (Kofax) redaction	Black out the same zone on every copy of a known form
DICOM pixel anonymisers (RSNA CTP, Orthanc Pixels Masker, XNAT)	Black out fixed rectangles per scanner type

Placeholder and restore for text

Work	What it does
HaS (Hide and Seek, 2023) and Portcullis (AAAI)	Hide sensitive text before a cloud model and restore it afterwards
Microsoft Presidio anonymize and deanonymize	Reversible operators that restore original values in the response
LLM Guard	The Anonymize scanner stores originals in a vault; the Deanonymize scanner restores them
LangChain reversible anonymizer, Skyflow and Protecto vaults	The same token and restore pattern as products

Reversible protection inside the image

Work	What it does	Why Finaira does not need it
Gu et al., "Password-conditioned Anonymization and Deanonymization with Face Identity Transformers", ECCV 2020	Replaces faces; the right password restores them	The crop never leaves the organization, so no cryptographic restore is needed
Reversible data hiding in encrypted images	Encrypts regions and restores them with a key	Same reason
CloakDiff (ACM MM 2026)	Reversible image protection against vision language models	Same reason

Risk to cite

Lopresti and Spitz, "Information Leakage Through Document Redaction: Attacks and Countermeasures", SPIE 2005. Redacted text was recovered from the size of the black box and leftover letter traces. Consequences for Finaira:

- placeholders must not reveal crop size or length;
- zones should be expanded slightly so no stray ink remains;
- a leftover ink check is needed.

The gap

ODK Scan and Shreddr crop by layout but send crops to people. Presidio, HaS and Portcullis restore placeholders but work on text only. No work was found that crops handwritten fields by layout, replaces them with typed placeholders, sends the document to a cloud language model, and pastes the crops back into the answer.

The "how it looks" column is based on the public blank bank forms collected (NBE, Banque Misr, HSBC) and on typical practice, not on filled real world copies.

References

- ODK-X Scan
- ODK Scan user guide
- Dell et al., GI 2013
- Shreddr, ACM DEV 2012
- ScaleHub snippeting
- IBM Datacap
- Patent US 2015/0213283
- Laserfiche Fixed Annotation
- Hyland OnBase
- Tungsten redaction
- Orthanc Pixels Masker
- XNAT pixel anonymization
- RSNA CTP
- HaS
- Portcullis
- Presidio deanonymize
- LLM Guard example
- LangChain data anonymizer
- Gu et al., ECCV 2020
- Reversible data hiding in encrypted images
- CloakDiff
- Lopresti and Spitz, SPIE 2005